Group Project: Portfolio Construction and Optimisation using Market Data

Introduction

The project is motivated by asset allocation and performance benchmarking in the Australian superannuation sector. APRA's MySuper Heatmap methodology (see pdf in Modules on Canvas) evaluates investment outcomes relative to benchmark portfolios that reflect a product's Strategic Asset Allocation (SAA) and its allocation to growth and defensive assets.

In this project, you will work as part of an investment team developing and evaluating an SAA model for a diversified superannuation portfolio. You will use historical Bloomberg index data for a range of asset classes to construct strategic portfolios with different growth/defensive allocations, evaluate their performance, and investigate whether tactical deviations from the strategic allocation can improve portfolio outcomes.

The project is deliberately open-ended: you will need to make and justify choices concerning the data, estimation period, portfolio constraints and numerical methods.

Let R denote the vector of asset returns, with

$$m = E[R], \qquad C = \mathrm{Cov}(R),$$

and let $x = (x_1, \ldots, x_n)^\top$ denote portfolio weights, with $\mathbf{1}^\top x = 1$. The parameter $\gamma > 0$ denotes risk aversion.

Parameter Estimation

Question 1. As a member of the investment team, you have been asked to estimate the inputs required for constructing a Strategic Asset Allocation (SAA) portfolio for a diversified investment fund, using the supplied Bloomberg data (`Bloomberg_data.xlsx` in Modules on Canvas). The data contain historical index levels for a range of asset classes. Note that the dates reported in the dataset do not necessarily correspond to the frequency at which new information becomes available for every asset.

(a) Investigate the available history and effective observation frequency of the different assets. Identify missing or repeated observations and determine whether observations appearing at daily or weekly dates always represent genuinely new information.

(b) Construct suitable return series. Choose and justify the data frequency and historical period used to estimate expected returns, volatilities and correlations. You may use a common frequency for all assets, or different frequencies where appropriate.

(c) Estimate annualised expected returns m and a variance–covariance matrix C. If different frequencies are used, explain how the estimates are converted to a common time scale and combined.

(d) Verify that your estimated C is a valid variance–covariance matrix. In particular, check that it is symmetric and positive semidefinite. If it is not, explain why this may have occurred and how you address the problem.

(e) Investigate how stable your parameter estimates are over time, for example using different estimation periods or rolling windows. Comment on the implications for portfolio construction.

Static Portfolio Optimisation

Question 2. Consider a Strategic Asset Allocation portfolio based on your parameter estimates from Question 1.

(a) Choose appropriate constraints on the portfolio weights, including limits on exposure to individual assets and whether short selling is allowed. Find and plot the efficient frontier.

(b) Find the global minimum-variance portfolio and report its allocation, expected return and volatility.

(c) For a suitable range of target returns μ, solve

$$\begin{aligned} \min_{x} \quad & x^\top C x, \\ \text{subject to} \quad & x^\top m \geq \mu, \qquad \mathbf{1}^\top x = 1, \end{aligned}$$

together with your other portfolio constraints.

(d) Compare the results obtained using different parameter estimates from Question 1. Comment on the sensitivity of the optimal portfolios to parameter estimation.

Question 3. Using the growth/defensive asset classification in the APRA MySuper Heatmap methodology, determine the growth and defensive classification of the assets in your dataset. Consider strategic allocations with total growth/defensive exposures of 30%/70%, 70%/30% and 90%/10%, which we refer to as defensive, balanced and aggressive portfolios.

(a) Find the minimum-variance portfolio for each allocation and compare their expected returns, volatilities and Sharpe ratios using an appropriately chosen risk-free rate.

(b) Discuss the diversification benefits of combining the different asset classes.

(c) Compare the three strategic allocations and discuss the circumstances under which each may be appropriate for an investment fund.

Utility and Monte Carlo Optimisation

Question 4. Consider an investor with initial wealth $W_0 = 1$ and exponential utility

$$U(W) = -e^{-\gamma W}, \qquad \gamma > 0.$$

For a one-period investment,

$$W_1 = 1 + x^\top R.$$

Assume first that $R \sim N(m, C)$. As shown in the tutorials, maximising expected utility is then equivalent to

$$\max_{x} \left\{ x^\top m - \frac{\gamma}{2} x^\top C x \right\}.$$

(a) Using your parameter estimates and portfolio constraints from the previous questions, find the optimal portfolio for several suitably chosen values of γ. Investigate how risk aversion affects the portfolio allocation, expected return and risk.

(b) Solve the same expected utility maximisation problem numerically under the normal return model, using your parameter estimates from Question 1. You may use Monte Carlo simulation to evaluate expected utility. Compare your numerical solution with the mean–variance benchmark obtained in part (a), and investigate the accuracy and computational performance of your numerical approach.

(c) Use a historical bootstrap to investigate the sampling uncertainty associated with the optimal portfolio in part (a). Investigate the resulting uncertainty in the portfolio allocation and discuss what this reveals about the reliability of the estimated optimal portfolio.

Out-of-Sample and Tactical Portfolio Allocation

Question 5. Investigate the out-of-sample performance of the defensive, balanced and aggressive allocation strategies considered in Question 3.

(a) Choose a historical date T_0 and an estimation window. Using only information available at or before T_0, re-estimate the model parameters and construct the defensive, balanced and aggressive portfolios using the same growth/defensive constraints as in Question 3.

(b) Keep the resulting portfolio weights fixed over a subsequent out-of-sample period. Compare the predicted and realised performance of the defensive, balanced and aggressive strategies using suitable measures of return and risk.

(c) Repeat the analysis for several choices of T_0. Comment on the stability of the portfolio allocations and their out-of-sample performance, and on the importance of avoiding look-ahead bias.

Question 6. Consider a Tactical Asset Allocation (TAA) strategy in which portfolio weights may be adjusted periodically in response to changing market information, while remaining within suitable limits around a chosen Strategic Asset Allocation (SAA).

Choose one of the strategic allocations considered in Questions 3 and 5 as the long-term SAA. At each rebalancing date T_k, use only information available at T_k to update your parameter estimates and determine the portfolio allocation.

(a) Specify and justify suitable limits on tactical deviations from the chosen SAA. Choose an estimation window, rebalancing frequency and value of the risk-aversion parameter γ.

(b) At each rebalancing date, determine the portfolio that maximises

$$x^\top m_k - \frac{\gamma}{2} x^\top C_k x$$

subject to your portfolio constraints and tactical allocation limits. Implement the strategy out of sample, using only information available at each rebalancing date.

(c) Compare the performance of the tactical strategy with that of the corresponding fixed strategic allocation from Question 5. Use cumulative wealth and suitable measures of return and risk.

(d) Investigate how the portfolio weights change over time and relate important changes to changes in the estimated expected returns, volatilities and correlations.

(e) Investigate the sensitivity of your results to at least one modelling choice, such as the estimation window, rebalancing frequency, tactical allocation limits or risk-aversion parameter.

(f) Does tactical reallocation add value relative to maintaining the fixed strategic allocation? Discuss your findings, including the effects of parameter estimation and portfolio turnover.

Extensions

You may undertake **at most one of the following extensions**. The **core questions are sufficient for a maximum mark of 85**. An extension should only be attempted after completing the core investigation.

Extension A: Advanced risk modelling and forecasting (maximum mark 100)

Extend the baseline portfolio model by investigating both the distribution and forecasting of portfolio risk.

Investigate whether the normal model adequately describes portfolio returns. Compare the normal model with alternative approaches, such as a Student-t model and the empirical distribution, and examine the implications for tail-risk measures such as Value-at-Risk (VaR) and Expected Shortfall (ES).

Develop a relatively simple statistical or machine-learning method to forecast a quantity relevant to portfolio risk or tactical asset allocation, such as volatility, correlations or expected returns. Use your forecasts to construct forward-looking measures of portfolio risk and incorporate them into the tactical portfolio strategy.

Evaluate both the risk forecasts and the resulting portfolio strategy out of sample. Compare with appropriate benchmarks and investigate whether the more advanced modelling and forecasting approach improves the assessment of portfolio risk or portfolio performance.

The emphasis should be on a careful numerical and out-of-sample investigation rather than on the complexity of the forecasting method. Discuss limitations such as estimation uncertainty, sample size and overfitting.

Extension B: Multiperiod dynamic portfolio optimisation (maximum mark 100)

Consider a multiperiod investor with

$$W_{k+1} = W_k(1 + x_k^\top R_{k+1}), \qquad k = 0, \ldots, N-1,$$

who maximises

$$E[U(W_N)], \qquad U(W) = -e^{-\gamma W}.$$

(a) Explain the difference between a fixed portfolio, the tactical strategy in Question 6, and a dynamic portfolio policy in which the allocation may depend on the investor's current wealth.

(b) Formulate the dynamic portfolio optimisation problem using dynamic programming. Define an appropriate value function and derive the corresponding Bellman recursion.

(c) Under a normally distributed return model, investigate the optimisation problem at the final investment date. Determine how the optimal allocation depends on current wealth and explain the financial interpretation of this dependence.

(d) For a small number of investment periods, develop and implement a numerical method for solving the dynamic optimisation problem. You may use Monte Carlo simulation and a discretisation of the wealth variable. Explain and justify your numerical choices and investigate the accuracy and stability of your method.

(e) Use your market data to calibrate the model and compare the resulting dynamic strategy with the fixed and tactical strategies considered earlier. Discuss when and why the strategies produce different portfolio allocations and performance.

Report and Presentation

Your report should be a self-contained document that tells a coherent story rather than a sequence of answers to the questions. Explain your modelling choices, methods and main mathematical ideas, and interpret your numerical results. Use appropriately chosen figures and tables. Do not include code in the main report; you may be asked to provide it separately. The report should be approximately 8–10 pages, excluding references. Groups undertaking an extension may use up to 12 pages. The emphasis should be on the most important results and conclusions rather than documenting every calculation or numerical experiment.

Further details about the requirements and assessment criteria for the group report and presentation will be provided later in the semester.

Students are required to present and demonstrate their interdisciplinary group project as a group in a mini-conference format. The mini-conference will be held on **Tuesday, 3 November (Week 13)**.

Presentation slides must be submitted via Canvas one day before the presentation. **Group project presentations** can only be delivered at the mini-conference on Tuesday, 3 November.

Students are required to prepare a **group report** summarising the outcomes of their interdisciplinary group project. The report must be submitted via Canvas one day before the presentation.

The group project report and presentation slides are due on **Monday, 2 November**, one day before the mini-conference. As this is a group assessment with presentations taking place at a scheduled mini-conference, extensions are not available. Late submissions of the group project report or presentation slides are not accepted.